

PROBLEM SET 5

(Due Feb 24, 2023 @ 8:30 AM in Gradescope and Moodle)

This problem set is to be completed **individually** and submitted in **Excel**. No hardcopies are needed. You will submit your assignments via Moodle and attach the screenshots to the Gradescope. Make sure you start with a blank Excel workbook on your own computer. Solve each problem on a separate Spreadsheet in Excel; name the Spreadsheets as “Problem 1”, “Problem 2”, and “Problem 3”

Problem 1 (25 points total)

Consider the following function: $f(x) = 3\cos(x^{1/5}) - x^3$

- (10 points) Use Excel to generate a plot of $f(x)$ vs. x on the interval $x \in [0, 2]$. Let x increase at a step size of 0.1. Label the plot properly. Next, use the graphical method to determine the solution of the above equation. In the spreadsheet, mark two cells in **red** denoting the boundaries of the x values between which you would find the solution to the above equation.
- (10 points) Use the fixed-point iteration method to find the solution to the above equation for $x > 0$. Here you will have to rearrange the equation and work on the iteration equation: $x_{i+1} = g(x_i)$. Provide the result with accuracy better than 1×10^{-7} . Mark your result in **red** on the spreadsheet.
- (5 points) Use Solver in Excel to find the root to the above equation for $x > 0$. Mark your result in **red** on the spreadsheet.

Problem 2 (35 points total)

Consider the function $f(x) = 2 - 9x + 7x^2 - x^3$ on the interval $x \in [0, 2]$; Let x increase at a step size of 0.1.

- (10 points) Use the graphical method to determine the solution(s) of the above equation. In the spreadsheet, mark the cells in **red**, denoting the boundaries of the x values between which you would expect to find the solution(s) to the above equation.
- (15 points) Determine the solution(s) of the above equation using the Newton-Raphson method. Mark your result in **red** on the spreadsheet.
- (10 points) Determine the solution(s) of the above equation using Solver in Excel. Mark your result in **red** on the spreadsheet.

Problem 3 (40 points total)

The Redlich-Kwong equation of state (RK-EOS) is frequently used to relate the gas pressure (P), volume (V), temperature (T) and the number of moles (n) for real gases. The equation reads as:

$$P = \frac{RT}{\frac{V}{n} - b} - \frac{a}{\sqrt{T} \frac{V}{n} (\frac{V}{n} + b)}.$$

The constants a and b are characteristic values for each gas and are given by:

$$a = 0.42780 \frac{R^2 \cdot T_C^{2.5}}{P_C},$$

$$b = 0.086640 \frac{R \cdot T_C}{P_C},$$

where T_C and P_C are the critical temperature and the critical pressure (see Table B1 in Felder, Rousseau & Bullard), respectively; R is the gas constant (8.314 J/(K*mol)). We want to calculate the gas volume (V) by using the RK-EOS in conjunction with the Newton-Raphson method and Solver in Excel.

- a) (10 points) Provide the iteration equation for V that would facilitate the Newton-Raphson method. Specify the expressions of $f(V)$ and $f'(V)$. The iteration equation should have the following form:

$$V_{i+1} = V_i - \dots$$

Type the necessary equations either directly into the Excel spreadsheet (Insert > Equation) or into a separate Word file and copy the equation into the excel.

- b) (10 points) Use the above derived equations to calculate the volume(s) of 2.0 moles of **oxygen** at $T=100^\circ\text{C}$ and under two different pressures: 1.0 kPa and 1.0 MPa. Supply the results with a precision better than 10^{-7} m^3 . Mark your result in **red** on the spreadsheet.
- c) (10 points) Find the solution to the above problems using Solver in Excel. Mark your result in **red** on the spreadsheet.
- d) (10 points) Calculate the percent error for both cases if the ideal gas equation-of-state (IG EOS) was instead used to calculate V. Use the following formula:

$$\% \text{ err} = \frac{|V_{RK} - V_{ideal}|}{V_{RK}} \times 100\%$$

Mark your result in **red** on the spreadsheet.